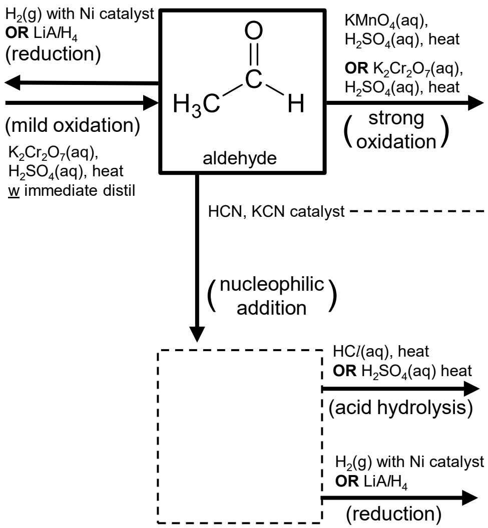
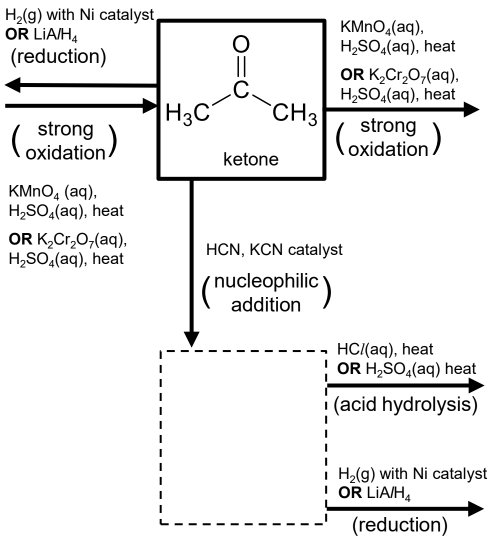
Reactions





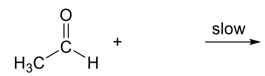
X

Mechanism

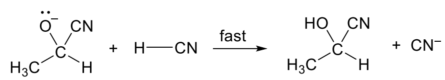
Name:

$KCN \longrightarrow K^+ + CN^-$

Step 1:



Step 2:



➢ Why is a racemic mixture formed?

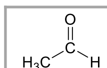
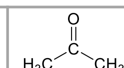
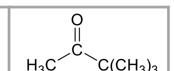
In step 1, the CN^- nucleophile can attack the ________________ carbonyl C from either side of the plane, at equal probability, hence forming the pair of enantiomers with equal proportions

➢ Why do carbonyls attract nucleophiles?

The carbonyl C is _________ as it is bonded to a more electonegative oxygen atom, therefore electron-rich nucleophiles are attracted to this _______________________________ site

Reactivity

Rank the following carbonyls from most to least reactive towards nucleophilic addition

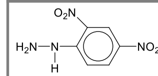
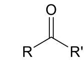
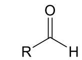
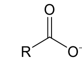
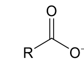
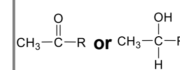
❶ **Electronic factor:**

The more electron-___________ (   ) the carbonyl C, the more susceptible it is to attack by nucleophiles.

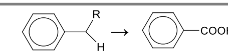
❷ **Steric factor:**

The ___________ the steric hindrance around the carbonyl C, the less susceptible it is to attack by nucleophiles.

Distinguishing tests

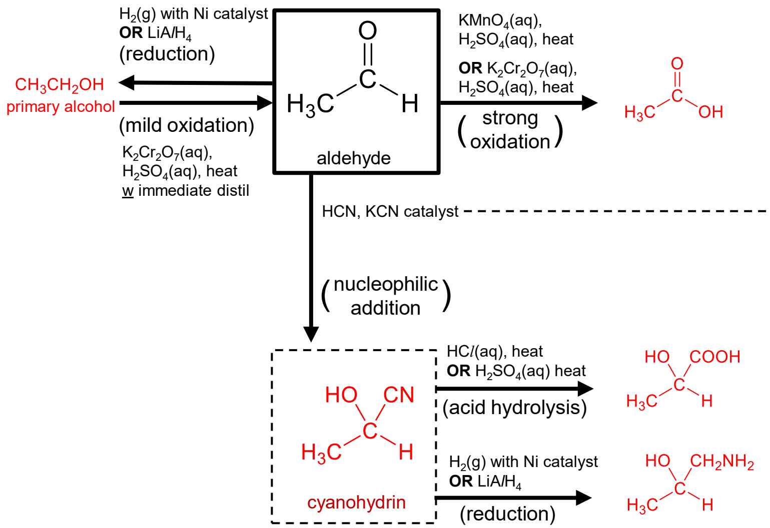
reagents / condn	type of rxn	test for	products formed
 2,4-DNPH		 aldehyde/ ketone	
Tollen's reagent, warm		 aldehyde	Ag silver mirror 
Fehling's solution, warm		only aliphatic aldehyde	Cu_2O brick red ppt 
NaOH(aq), I_2(aq), warm (iodoform test)			

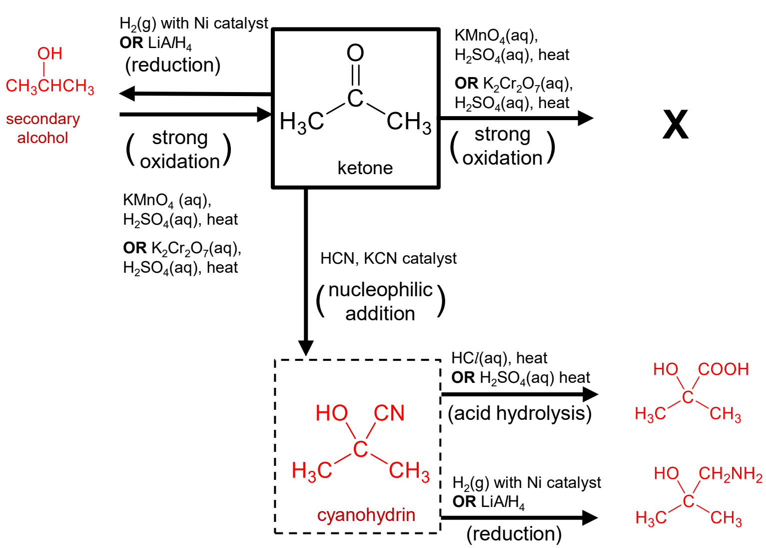
Oxidising & reducing agents

Oxidation	$KMnO_4$(aq), H_2SO_4(aq), heat	$K_2Cr_2O_7$(aq), H_2SO_4(aq), heat
alkene → RCOOH/ CO_2/ ketone	✓	
alkene → diol	✓ (NaOH(aq), cold)	
	✓	
1° alcohol → aldehyde		✓ (immediate distil)
2° alcohol → ketone	✓	✓
aldehyde → RCOOH	✓	✓

Reduction	H_2(g) with Ni catalyst	$LiAlH_4$
alkene → alkane	✓	
RC≡N → RCH_2NH_2	✓	✓
aldehyde → 1° alcohol	✓	✓
ketone → 2° alcohol	✓	✓
RCOOH → 1° alcohol		✓
$RCONH_2$ → RCH_2NH_2		✓

Reactions



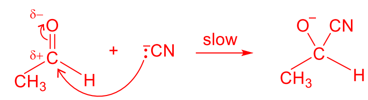


Mechanism

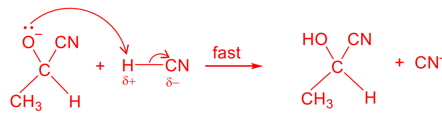
Name: nucleophilic addition

$KCN \longrightarrow K^+ + CN^-$

Step 1:



Step 2:



➢ Why is a racemic mixture formed?

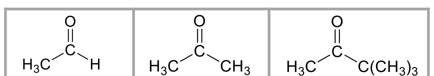
In step 1, the CN^- nucleophile can attack the sp^2 planar carbonyl C from either side of the plane, at equal probability, hence forming the pair of enantiomers with equal proportions

➢ Why do carbonyls attract nucleophiles?

The carbonyl C is $\delta+$ as it is bonded to a more electronegative oxygen atom, therefore electron-rich nucleophiles are attracted to this electron-deficient site

Reactivity

Rank the following carbonyls from most to least reactive towards nucleophilic addition



most reactive → least reactive

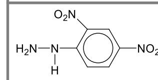
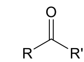
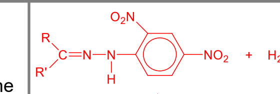
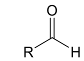
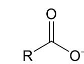
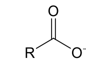
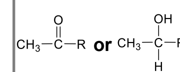
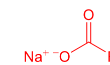
❶ **Electronic factor:**

The more electron-deficient ($\delta+$) the carbonyl C, the more susceptible it is to attack by nucleophiles.

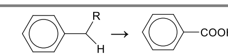
❷ **Steric factor:**

The greater the steric hindrance around the carbonyl C, the less susceptible it is to attack by nucleophiles.

Distinguishing tests

reagents / cond^n	type of rxn	test for	products formed
 2,4-DNPH	condensation	 aldehyde/ ketone	 orange ppt
Tollen's reagent, warm	oxidation	 aldehyde	Ag silver mirror 
Fehling's solution, warm	oxidation	only aliphatic aldehyde	Cu_2O brick red ppt 
NaOH(aq), I_2(aq), warm (iodoform test)	oxidation		CHI_3 yellow ppt 

Oxidising & reducing agents

Oxidation	$KMnO_4$(aq), H_2SO_4(aq), heat	$K_2Cr_2O_7$(aq), H_2SO_4(aq), heat
alkene → RCOOH/ CO_2/ ketone	✓	
alkene → diol	✓ (NaOH(aq), cold)	
	✓	
1º alcohol → aldehyde		✓ (immediate distil)
2º alcohol → ketone	✓	✓
aldehyde → RCOOH	✓	✓

Reduction	H$_2$(g) with Ni catalyst	LiAlH$_4$
alkene → alkane	✓	
RC≡N → RCH_2NH_2	✓	✓
aldehyde → 1º alcohol	✓	✓
ketone → 2º alcohol	✓	✓
RCOOH → 1º alcohol		✓
$RCONH_2$ → RCH_2NH_2		✓